Deep Learning-Based Morphological and Microstructural Analysis of Porous Ceramics

Thesis submitted in partial fulfilment of the requirements for the award of

Integrated Dual Degree (Bachelor of Technology and Master of Technology)

In

CERAMIC ENGINEERING

Submitted by

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Under the guidance of

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MAY 2026

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I, **Saagar Nandini Surendra**, certify that the research expressed in this dissertation is my own work, which I have carried out, under the supervision of **Dr. Subrata Panda**, in the **Department of Ceramic Engineering, Indian Institute of Technology (BHU), Varanasi** from January 2025 to May 2026. The subject of this thesis was not submitted for the recognition of another degree / diploma. I acknowledge that I have sincerely accepted and given credit to the researchers wherever their work was cited in my thesis.

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It is certified that the above statement made by the student is correct to the best of my knowledge.

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This is to certify that the dissertation entitled **"Deep Learning-Based Morphological and Microstructural Analysis of Porous Ceramics"**, submitted by **Ms. Saagar Nandini Surendra** (Roll No. 21034011) in partial fulfilment for the award of **Integrated Dual Degree (Bachelor of Technology and Master of Technology)** in **Ceramic Engineering** to the **Indian Institute of Technology (BHU), Varanasi**, is a faithful record of Bona fide research carried out by herself under my guidance and supervision. To the best of my knowledge the matter embodied in this dissertation has not been submitted to any other University/Institute for the award of any other degree.

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Acknowledgements

I am proud to pursue my thesis work under the guidance of **Dr. Subrata Panda, Department of Ceramic Engineering, IIT(BHU) Varanasi**. No words are adequate to express my sincere gratitude towards him for guiding me through the project with his scholastic expertise and enriched experience, for being ready to assist me with any problem I face, and for his consistent encouragement and motivation not only for my project but also for my future career plans. I am grateful to have served under his guidance. My thankful regards are due to **Vivek Kumar Saroj, Md. Arif** and **Nikhil Devgan** for their valuable guidance during my research. I also would like to thank my lab members and batch mates for their support.

A heartfelt mention of gratitude to my parents, **Mrs. Kalpana Saagar** and **Mr. Surendra Saagar**, for being my strongest support system, providing quality education, and shaping me into who I am today.

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To

My hardworking father,
Mr. Surendra Saagar

Abstract

Porous ceramics are used in demanding engineering applications such as thermal insulation, filtration membranes, and biomedical scaffolds, where performance is governed directly by pore architecture: how much void space exists, how pores are sized and distributed, and how well they are connected. Conventional SEM analysis relies on manual or semi-automated thresholding, which breaks down under uneven illumination, inter-operator disagreement, and imaging noise, and does not scale to the image volumes needed for statistically meaningful characterisation.

This thesis develops and benchmarks a deep learning pipeline for automated pore segmentation and morphological characterisation of SEM images from seven porous ceramic specimens prepared at IIT (BHU), Varanasi. These specimens span a diverse range of pore morphologies, from isolated near-spherical pores to densely interconnected networks. Ground truth was generated by soft-blending an automated classical mask with a semi-manual ImageJ mask, preserving boundary uncertainty rather than forcing hard labels. CLAHE-preprocessed images were tiled into 256×256 patches, augmented five-fold, and used to train five architectures: SegNet, U-Net, DeepLabV3+, Attention U-Net, and TransUNet, under identical conditions on an NVIDIA T4 GPU.

Attention U-Net achieved the best test-set performance (Dice = 0.983, IoU = 0.967), with U-Net a close second (Dice = 0.982); both estimated specimen porosity within 0.2 percentage points of ground truth. DeepLabV3+, trained without ImageNet pretraining on a single-channel input, lagged behind at Dice = 0.869. Following segmentation, connected-component analysis extracted morphological descriptors for each specimen, including equivalent diameter percentiles, aspect ratio, solidity, and connectivity ratio. Above a critical porosity threshold, the connectivity ratio rises sharply, marking the transition to a percolating, channel-like pore network with direct implications for mechanical strength and transport behaviour.

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List of Abbreviations

Abbreviation	Full Form / Meaning
AG	Attention Gate
ASPP	Atrous Spatial Pyramid Pooling
BCE	Binary Cross-Entropy
CLAHE	Contrast Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CR	Connectivity Ratio
D10/D50/D90	10th / 50th / 90th percentile of equivalent-diameter distribution
Dice	Sørensen–Dice coefficient
FIB-SEM	Focused Ion Beam Scanning Electron Microscopy
GPU	Graphics Processing Unit
GT	Ground Truth
IoU	Intersection over Union
ML	Machine Learning
MLP	Multi-Layer Perceptron
PA	Pixel Accuracy
PSD	Pore Size Distribution
SEM	Scanning Electron Microscopy
TTA	Test-Time Augmentation
ViT	Vision Transformer

Chapter 1

Introduction

1.1 Background and Motivation

Porous ceramics constitute a class of advanced engineering materials characterised by a significant volume fraction of voids that may be open, closed, or interconnected, distributed throughout their solid matrix (*Figure 1.1*). This distinctive architecture directly governs the material's physical, mechanical, and thermal behaviour, making deliberate control of pore morphology indispensable for performance-critical applications. Due to their adjustable properties, porous ceramics are utilized in a wide range of technologies. These applications include thermal insulation systems, lightweight structural components, filtration membranes, catalytic support substrates, energy storage devices, and biomedical scaffolds for tissue engineering and drug delivery [1].

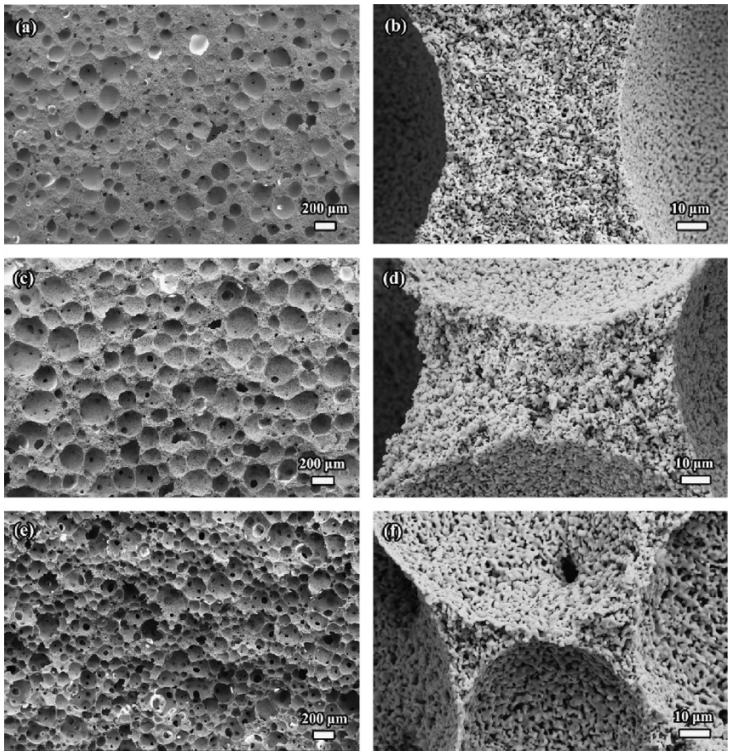


Figure 1.1: SEM micrographs of porous γ-$Y_2Si_2O_7$ ceramic at 64.3%, 77.2%, and 88.2% porosity. *From [2].*

The central engineering challenge posed by porous ceramics is the inherent trade-off between porosity and mechanical integrity: increasing the void fraction to enhance functional performance simultaneously diminishes compressive and flexural strength. Optimising this balance demands precise, quantitative characterisation of the pore network, specifically its porosity fraction, pore-size distribution, connectivity, shape descriptors, and relative density [1]. Scanning Electron Microscopy (SEM) has long served as the principal imaging modality for this purpose, providing high-resolution two-dimensional cross-sections that encode the full morphological richness of the pore structure.

1.2 Limitations of Traditional SEM-Based Analysis

Despite the considerable advantages offered by SEM, traditional image-analysis workflows are hindered by several well-documented limitations, which underscore the need for incorporating machine learning techniques. *Table 1.1* presents a summary of the four primary challenges identified in the existing literature [1].

Table 1.1: Principal challenges in SEM-based pore image analysis. Summarised from [1].

Challenge	Description	Impact on Analysis
Noise & Artefacts	Shot noise from limited electrons per pixel; charging effects and beam-induced shadowing further distort images.	Obscures fine pore boundaries; reduces segmentation accuracy.
Segmentation Difficulty	Irregular pore shapes with varying size, orientation, and interconnection defeat conventional intensity-thresholding methods.	Misclassification of pore boundaries; inaccurate size and shape quantification.
Resolution Limits	Trade-off between high resolution and beam-induced sample damage; sub-resolution pores require indirect estimation.	Sub-micron pores undetectable; reduced precision for delicate materials.
Subjectivity	Human analysts apply inconsistent thresholds and boundary criteria across large datasets.	Poor reproducibility; labour-intensive; unsuitable for high-throughput studies.

These limitations collectively undermine the scalability, reproducibility, and accuracy of microstructural characterisation at the volumes required for statistically robust structure–

property inference, providing the primary motivation for integrating machine learning into the SEM analysis pipeline.

1.3 Deep Learning for Microstructural Analysis

The application of machine learning to SEM-based microstructure analysis has advanced rapidly over the past decade, driven by the capacity of deep neural networks to learn discriminative, hierarchical representations directly from image data. As reviewed comprehensively in [1], ML methods now address four complementary tasks in the pore-analysis workflow, illustrated schematically in *Figure 1.2*.

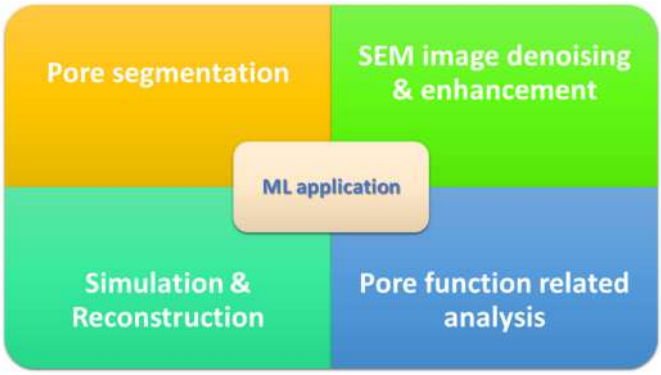


Figure 1.2: Four application areas of ML in SEM-based pore analysis. *Reproduced from [1].*

The fully integrated AI SEM pipeline, covering all four stages from raw image acquisition to quantitative pore-property extraction, is illustrated in *Figure 1.3*.

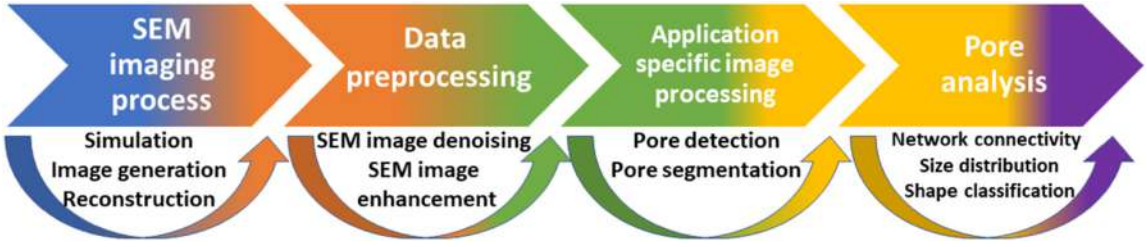


Figure 1.3: AI-integrated SEM pore-analysis pipeline from image acquisition to quantitative characterisation. *Reproduced from [1].*

Within this pipeline, semantic segmentation, the pixel-level classification of pore versus matrix regions, is the most foundational task, as every downstream morphological descriptor (porosity fraction, pore-size distribution, shape factor, connectivity) depends directly on its accuracy. The literature has converged on two broad architectural families for this task:

(i) CNN-based models such as SegNet, U-Net, DeepLabV3+, and Attention U-Net, which exploit hierarchical local spatial features and, in encoder–decoder designs, preserve fine boundary detail through skip connections; and

(ii) Transformer-based hybrid models such as TransUNet, which augment convolutional feature extraction with self-attention mechanisms to capture long-range spatial dependencies that purely local convolutions cannot model.

Deep-learning models in these families achieve pore-detection accuracies reported up to 97% on ceramic and related porous-material datasets [1], while some also support three-dimensional reconstruction from single two-dimensional images. In the domain of ceramic microstructures specifically, U-Net architectures have demonstrated high-fidelity segmentation of three-dimensional SEM volumes, capturing defect structures characteristic of polycrystalline ceramics [9].

1.4 Objectives of the Study

The primary objective of this thesis is:

> *To model an optimised porous ceramic microstructure through the development, training, and systematic benchmarking of five deep-learning image segmentation architectures applied to Scanning Electron Microscopy (SEM) images, to achieve automated, accurate, and reproducible pixel-wise classification between pore and matrix regions as a prerequisite for downstream structure–property prediction.*

Concretely, this entails acquiring and pre-processing an annotated SEM dataset of porous ceramic specimens, implementing five architectures that span the full spectrum of modern segmentation design from a lightweight encoder–decoder with no skip connections to a hybrid CNN-Vision Transformer, and evaluating each under a unified training and evaluation protocol using pixel-level accuracy and related metrics. The comparison is intended not merely to identify the highest-performing model, but to provide a principled, evidence-based understanding of *which architectural features* (skip connections, multi-scale pooling, attention gating, global self-attention) drive performance gains. The context is porous ceramic SEM imagery, where pores vary widely in size, shape, and contrast.

The five models are evaluated in order of increasing architectural complexity, as summarised in *Table 1.2*.

Table 1.2: Five segmentation architectures benchmarked in this study, ordered by increasing complexity.

No.	Model	Key Architectural Feature
1	**SegNet**	Encoder–decoder with pooling-index upsampling; no skip connections [3]
2	**U-Net**	Symmetric skip connections preserving spatial boundary detail [4]
3	**DeepLabV3+**	Atrous Spatial Pyramid Pooling (ASPP) for multi-scale context [5]
4	**Attention U-Net**	Soft Attention Gates suppressing irrelevant background activations [6]
5	**TransUNet**	Hybrid CNN + Vision Transformer capturing global spatial dependencies [8]

1. **SegNet** [3] serves as the simplest baseline model. Its encoder–decoder architecture reuses max-pooling indices from the encoder during upsampling, achieving memory-efficient segmentation without skip connections. This establishes the lower-bound performance reference against which all subsequent architectures are evaluated.

2. **U-Net** [4] introduces symmetric skip connections that concatenate encoder feature maps directly into corresponding decoder layers, preserving high-frequency spatial information critical for accurate pore boundary delineation. It represents the standard reference architecture in materials microstructure segmentation and has been widely validated on ceramic SEM data [9].

3. **DeepLabV3+** [5] employs Atrous Spatial Pyramid Pooling (ASPP) to simultaneously aggregate multi-scale contextual information at multiple receptive field sizes, followed by a lightweight decoder that refines boundary localisation. This is particularly advantageous for porous ceramics, where pores span a wide range of spatial scales.

4. **Attention U-Net** [6] augments the U-Net architecture with soft Attention Gates (AGs) at every skip connection. Each gate generates a gating signal that dynamically suppresses irrelevant background activations while amplifying salient pore-boundary

features, improving recall for small and low-contrast pores with negligible additional computational overhead [7].

5. **TransUNet** [8] is the most sophisticated model evaluated. It hybridises a CNN encoder with a Vision Transformer (ViT) module, which tokenises intermediate feature maps and applies multi-head self-attention to model global spatial relationships across the full image. This enables the model to reason about long-range pore inter-connectivity that purely convolutional operations cannot capture.

These five models span the full range from a lightweight CNN baseline to a hybrid CNN-Transformer, giving a clear picture of where each design choice helps and where it does not on this class of materials imagery.

1.5 Significance and Relevance

The relevance of this work spans three intersecting domains. From a **materials engineering** perspective, automated and reproducible segmentation of pore networks is a prerequisite for reliable property prediction and process optimisation. Manual analysis of SEM datasets is prohibitively time-consuming for the sample sizes required by statistical inference; a high-accuracy deep learning pipeline eliminates this bottleneck entirely. From a **computational** perspective, the systematic five-model comparison provides an evidence-based guide for architecture selection in materials characterisation tasks where annotated training data are scarce, and boundary precision is essentially a contribution of broad value to the materials informatics community. From an **applications** perspective, the morphological descriptors extracted by this pipeline can be directly correlated with macroscopic properties (compressive strength, thermal conductivity, gas permeability) without manual intervention, enabling analysis of hundreds of micrographs in hours rather than weeks. This thesis contributes a multi-architecture segmentation benchmark on ceramic SEM data, providing empirical evidence for which architectural families generalise well under small-dataset constraints.

1.6 Scope and Limitations

Scope

This thesis is scoped to the segmentation stage of the broader DL-based porous ceramic characterisation pipeline. The work encompasses:

- Collection and annotation of SEM images of porous ceramic specimens representing a range of pore morphologies, sizes, and distributions.
- Pre-processing and augmentation of the SEM dataset to address class imbalance between pore and matrix pixels and to improve model generalisation.
- Implementation and training of five deep-learning segmentation architectures: SegNet, U-Net, DeepLab V3+, Attention U-Net, and TransUNet under a unified experimental protocol.
- Quantitative benchmarking of all five models using pixel-level accuracy and auxiliary segmentation metrics, followed by qualitative analysis of predicted segmentation maps.
- Identification of the best-performing architecture and a principled explanation of the architectural features responsible for its superior performance on porous ceramic microstructures.

Limitations

Several limitations bound the conclusions drawn in this thesis and define natural directions for future work:

1. **Single material system.** The models are trained and evaluated exclusively on porous ceramic SEM images. Generalisation to other porous materials (e.g. metallic foams, rock cores, polymer scaffolds) is not investigated and cannot be assumed without further validation.
2. **Two-dimensional imaging.** All SEM images used in this study are two-dimensional cross-sections. Three-dimensional pore connectivity, tortuosity, and through-thickness porosity gradients are therefore not captured. Extension to FIB-SEM volumetric stacks is identified as future work.

3. **Dataset size and diversity.** The volume and compositional diversity of the annotated dataset are constrained by the availability of labelled SEM images and the cost of expert annotation. Larger, more diverse datasets may yield different relative rankings among the five architectures.

4. **Segmentation only.** This thesis does not address the downstream prediction of thermal or mechanical properties from the segmented microstructures. The segmentation outputs produced here serve as the necessary foundation for that subsequent work.

5. **Hyperparameter optimisation.** Systematic hyperparameter search (e.g. neural architecture search or Bayesian optimisation) is not performed; standard choices from the respective original publications are adopted with task-specific adjustments, which may not represent the absolute optimal configurations for the porous ceramic domain.

Chapter 2

Literature Review

2.1 Porous Ceramic Characterisation

Porous ceramics such as alumina (Al_2O_3), zirconia (ZrO_2), silicon carbide (SiC), and yttrium disilicate (γ-$Y_2Si_2O_7$) are manufactured via replica templating, direct foaming, freeze-casting, and sacrificial-filler methods [12, 13]. Each route produces characteristic pore topologies that must be quantified to predict functional behaviour.

2.1.1 Key Morphological Descriptors

Table 2.1 summarises the key descriptors extracted from SEM images. These descriptors are used as input features for subsequent property prediction models. The limitations associated with manual descriptor extraction are discussed in Section 1.2 of Chapter 1.

Table 2.1: Key morphological descriptors of porous ceramics and their functional relevance.

Descriptor	Definition	Functional Relevance
Porosity fraction (ϕ)	Ratio of void area to total image area	Inversely correlates with compressive strength
Pore-size distribution	Histogram of equivalent circular diameters	Controls filtration cut-off and permeability
Shape factor (circularity)	$4\pi A/P^2$; area A, perimeter P	Stress concentration; thermal shock resistance
Connectivity / tortuosity	Path-length ratio through pore network	Ion and fluid transport efficiency
Relative density	$1 - \phi$ (solid fraction)	Elastic modulus (Gibson–Ashby scaling)
Specific surface area	Pore-boundary length per unit area	Catalytic activity; drug-loading capacity

2.1.2 Structure–Property Relationships

Mechanical and thermal properties depend strongly and non-linearly on pore architecture. The Gibson–Ashby power-law [14] relates compressive strength to relative density:

$$\frac{\sigma_c}{\sigma_s} = C_1 \left(\frac{\rho^*}{\rho_s} \right)^n, \qquad n \approx 1.5 \text{ (open-cell)}, \;\; 2 \text{ (closed-cell)} \tag{2.1}$$

where σ_s and ρ_s are the strength and density of the fully dense matrix. Because the exponent n amplifies porosity errors, even a small segmentation mistake propagates into a measurable strength prediction error. Thermal conductivity follows a similar pattern: effective-medium models are sensitive to pore connectivity, so distinguishing open from closed pore populations matters for reliable property estimates [1]. Both cases point to the same requirement: segmentation must be accurate at the individual-pore level, not just in aggregate porosity fraction.

2.1.3 Conventional SEM Analysis and Its Failure Modes

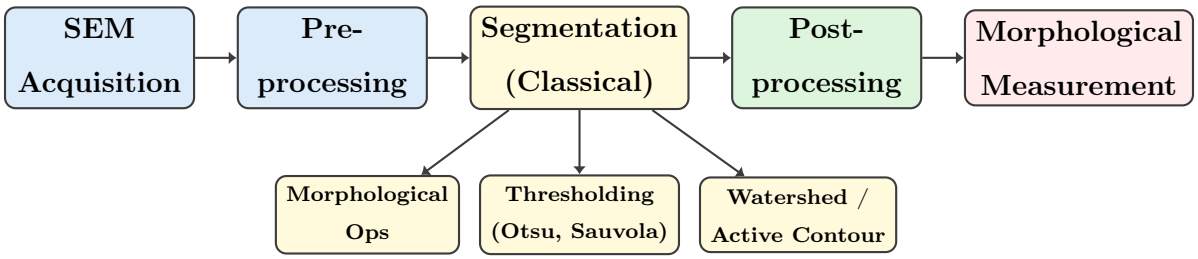


Figure 2.1: Conventional SEM pore-segmentation pipeline with three classical thresholding strategies.

The limitations of conventional SEM-based workflows, namely noise sensitivity, operator variability, and poor scalability, were catalogued in *Table 1.1*. These motivate a shift toward learned segmentation, where the model adapts its feature extraction to the specific contrast and geometry of each image rather than relying on fixed thresholds.

2.2 Deep Learning in SEM Pore Analysis

2.2.1 Functional Areas of ML-Based Microstructural Analysis

The principal categories of machine learning applications in SEM-based pore analysis are illustrated in *Figure 2.2*, building upon the conceptual overview presented earlier

in Chapter 1 by highlighting representative models and techniques used within each application domain.

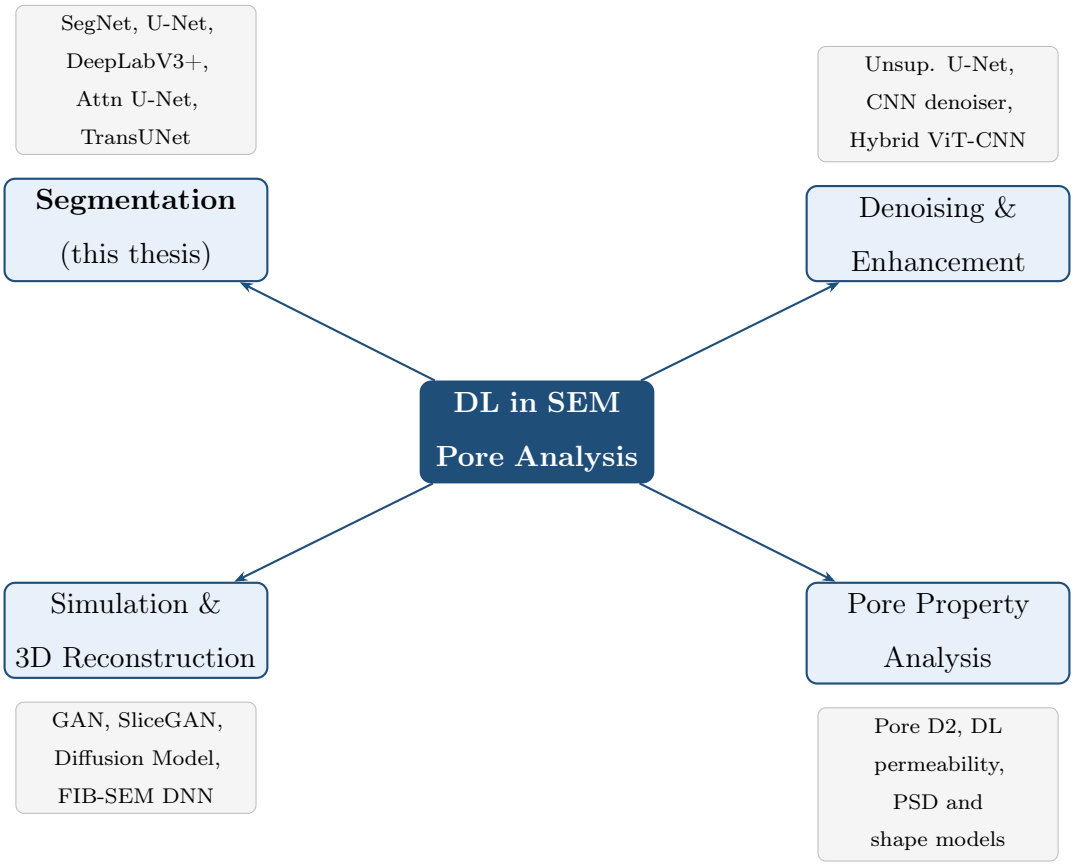


Figure 2.2: ML application taxonomy for SEM pore analysis; this thesis addresses the segmentation quadrant. Adapted from [1].

2.2.2 Denoising, Property Analysis, and 3D Reconstruction

Denoising: Unsupervised U-Net frameworks achieve effective contour extraction under low signal-to-noise conditions without requiring clean reference images, critical for ceramic specimens where beam damage limits exposure time [1]. Hybrid ViT-CNN networks extend this by jointly handling noise attenuation and boundary preservation within a single model.

Property analysis: Deep learning models predict permeability from 2D SEM images in seconds, replacing costly Lattice Boltzmann simulations [1]. Diffusion Probabilistic Models outperform GANs in preserving topological fidelity of porous copper microstructures.

3D reconstruction: SliceGAN generates high-fidelity 3D volumes from a single 2D image by enforcing statistical isotropy through adversarial training [1]. GAN-based 3D

reconstruction of porous ceramic electrodes shows statistically consistent pore network topology against FIB-SEM ground truth [17].

2.3 Review of Segmentation Architecture

Figure 2.3 maps the five architectures onto axes of boundary precision and spatial context, showing the incremental capability gain from SegNet to TransUNet.

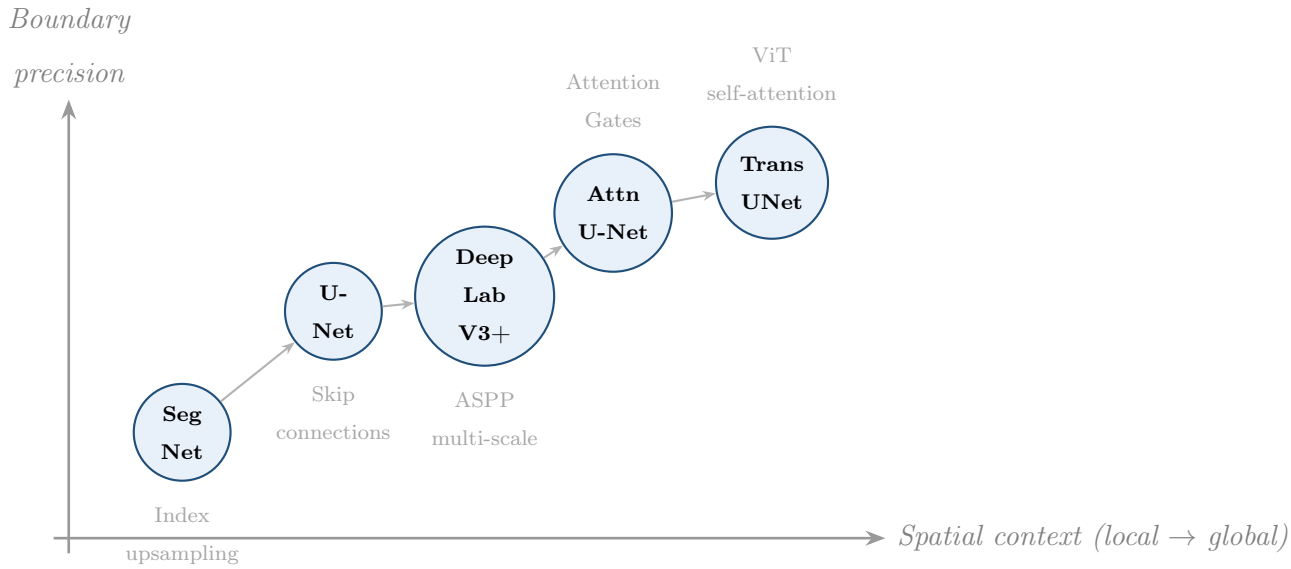


Figure 2.3: Architectural evolution from SegNet (2017) to TransUNet (2021).

2.3.1 SegNet

Badrinarayanan *et al.* [3] introduced SegNet (2017) as a parameter-efficient encoder–decoder architecture employing a VGG-16 encoder and pooling-index–based upsampling within the decoder. Unlike U-Net–style architectures, encoder feature maps are not directly transferred to the decoder, limiting the recovery of fine boundary information lost during downsampling. Comparative studies on steel SEM microstructures [15] demonstrated that SegNet produced lower IoU performance than both U-Net and DeepLabV3+, establishing SegNet primarily as a baseline reference architecture for segmentation benchmarking.

2.3.2 U-Net

Ronneberger *et al.* [4] introduced U-Net (2015) with full concatenative skip connections at every encoder resolution level. This allows the decoder to simultaneously access deep semantic features and shallow spatial features, directly addressing the boundary precision

limitation of SegNet. For 3D FIB-SEM volumes of polycrystalline ceramics, Hara *et al.* [9] confirmed that U-Net recognises grain boundaries, pores, and secondary phases with accuracy previously requiring expert manual annotation. On concrete SEM images, U-Net outperformed Otsu thresholding in accuracy and throughput [16]. U-Net is the primary reference in this thesis.

2.3.3 DeepLabV3+

Chen *et al.* [5] introduced *Atrous Spatial Pyramid Pooling* (ASPP) in DeepLabV3+ (2018), applying dilated convolutions at rates $r \in \{6, 12, 18\}$ in parallel to capture context at multiple spatial scales without resolution loss. This is directly motivated by the porous ceramic challenge: macro-pores (200 μm scale) and micro-pores (10 μm scale) coexist in the same specimen (*Figure 1.1*), and no single receptive field captures both. A direct comparison on steel SEM data confirmed the highest mIoU among CNN-only models [15].

2.3.4 Attention U-Net

Oktay *et al.* [6] introduced *soft Attention Gates* (AGs) at every U-Net skip connection (2018). Each gate computes an attention map $\alpha_i \in (0, 1)$ from the decoder gating signal and the encoder feature, suppressing background matrix activations and amplifying pore boundary features:

$$\hat{x}_i^l = \alpha_i \cdot x_i^l, \qquad \alpha_i = \sigma\big(\mathbf{W}_\psi \left[\sigma(\mathbf{W}_g g_i + \mathbf{W}_x x_i^l)\right]\big) \tag{2.2}$$

The gating operation is differentiable and adds negligible overhead [7]. Wang *et al.* [11] demonstrated superior sensitivity to small pores in sintered ceramic paste over standard U-Net; attention mechanisms also improved boundary delineation in low-contrast rock-core SEM images [10]. The **Attn** column in Table 3.3 records which architectures in this comparison employ this gating mechanism.

2.3.5 TransUNet

Chen *et al.* [8] proposed TransUNet (2021), the first architecture to integrate a Vision Transformer (ViT) encoder into the U-Net framework. *Figure 2.4* illustrates the two-stage encoding pipeline.

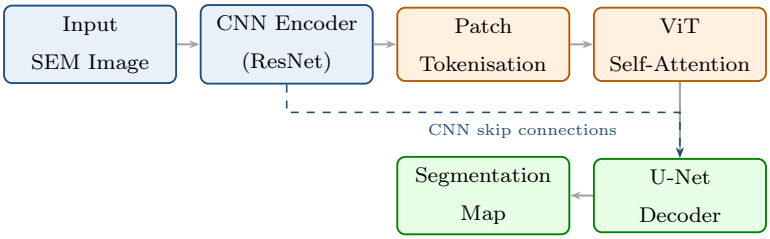


Figure 2.4: TransUNet: ResNet CNN encoder feeds a ViT for global context; a U-Net decoder reconstructs the segmentation map. Adapted from [8].

The ViT encoder computes self-attention across all patch tokens, enabling the model to capture long-range spatial relationships between pore regions that may belong to the same connected microstructural network. This capability allows transformer-based architectures to model global contextual information beyond the receptive field limitations of conventional convolutions [18]. Subsequent variants such as Swin-UNet, DS-TransUNet, and DA-TransUNet further demonstrate the growing importance of transformer-based attention mechanisms in complex image segmentation tasks [19].

Nevertheless, the advantages of global self-attention are accompanied by several practical limitations. Transformer-based encoders generally require substantially larger labelled datasets than CNN-based architectures to learn stable and meaningful attention representations [8]. This requirement can become restrictive in SEM-based ceramic pore analysis, where annotated datasets are often limited in size. In addition, when pore connectivity is governed predominantly by local structural variations rather than long-range interactions, the computational complexity introduced by transformer encoders may not provide proportional segmentation improvements over carefully optimised CNN attention mechanisms. Consequently, Attention U-Net may offer superior performance under data-constrained ceramic segmentation conditions, a proposition investigated experimentally in this thesis.

2.4 Comparative Summary

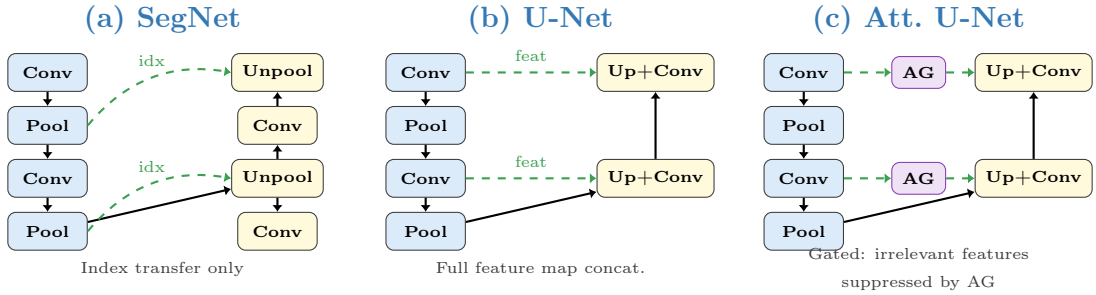


Figure 2.5: Skip-connection strategies: (a) SegNet pooling indices, (b) U-Net full feature concatenation, (c) Attention U-Net gated connections.

Table 2.2: Comparative summary of five architectures across key design dimensions for porous ceramic SEM segmentation.

Property	SegNet	U-Net	DeepLabV3+	Attn U-Net	TransUNet
Skip connections	Index only	Full concat.	Partial	Gated concat.	CNN skip
Multi-scale context	No	Limited	ASPP	Limited	ViT global
Attention	None	None	None	Soft gates	Self-attn
Long-range modelling	No	No	Partial	No	Yes
Parameters	Low	Medium	High	Medium+	High
Boundary precision	Moderate	Good	Good	Very good	Very good
Ceramic SEM use	Reference	Extensive	Growing	Growing	Emerging

Table 2.2 consolidates the five architectures across dimensions relevant to porous ceramic segmentation.

Each row captures a capability that generally increases from SegNet to TransUNet. Crucially, however, architectural complexity does not guarantee superior empirical performance: dataset size, pore connectivity patterns, and domain characteristics can cause a lighter, well-targeted model such as Attention U-Net to outperform a more complex one such as TransUNet. This nuance is central to the experimental contribution of this thesis.

2.5 Evaluation Metrics

All models are evaluated using a unified set of binary segmentation metrics [1, 9, 11]:

$$\text{PA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \qquad \text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \qquad \text{Dice} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}} \tag{2.3}$$

Dice is the primary metric used throughout this thesis; IoU is reported as a complementary overlap measure that is more sensitive to boundary errors. Pixel Accuracy (PA) is additionally reported for cross-study comparability. Dice corrects for class imbalance between minority pore and majority matrix pixels. Precision and Recall are additionally reported to distinguish false positive (phantom pore) and false negative (missed pore) error modes independently.

2.6 Research Gaps and Thesis Contribution

Four clear gaps emerge from this review:

1. **No systematic five-model benchmark on porous ceramics.** Existing studies test one or two architectures in isolation on a specific material [1, 9]. No study has benchmarked all five architectures on the same ceramic SEM dataset under a unified experimental protocol.

2. **Transformer models untested on inorganic porous materials.** TransUNet and successors have been validated solely in medical imaging. Porous ceramic microstructures differ fundamentally from biological tissue in contrast characteristics, pore geometry, and spatial scale. Whether global self-attention provides a net benefit over local attention-gating on ceramic SEM data remains an open question.

3. **Absence of architecturally interpretable comparisons.** Performance differences are rarely attributed to specific design features, preventing principled model selection for new material systems.

4. **Unverified assumption that architectural complexity implies superior accuracy.** Existing literature often assumes that increasingly complex segmentation architectures inherently produce superior performance. However, for small and domain-specific datasets such as annotated ceramic SEM micrographs, this assumption may not necessarily remain valid. Under data-constrained conditions, architectures employing

local attention mechanisms may achieve better generalisation than globally attentive transformer-based models, particularly when pore connectivity is dominated by local microstructural relationships rather than long-range spatial dependencies.

This thesis addresses all four gaps. In particular, the experimental finding that Attention U-Net outperforms TransUNet on this ceramic SEM dataset directly challenges the complexity-implies-accuracy assumption, providing a practically important and architecturally interpretable result for the materials characterisation community. The corresponding experimental validations are presented in Chapter 4.

Chapter 3

Methodology

3.1 Experimental Framework Overview

The pipeline comprises seven phases: dataset preparation, image preprocessing, ground-truth mask generation, patch extraction and augmentation, model training, inference, and pore characterisation. *Figure 3.1* shows the complete workflow. All five architectures are trained and evaluated under a single, unified protocol so that performance differences in Chapter 4 are attributable to architecture alone, not to differing data or training conditions.

The entire pipeline is implemented in Python 3.12 using PyTorch for all deep learning components, OpenCV and scikit-image for classical image processing, and SciPy for morphological operations and spatial analysis. Model training was executed on an NVIDIA T4 GPU (16 GB VRAM) via the Kaggle cloud environment. *Table 3.1* lists the software stack.

Table 3.1: Software and hardware environment.

Component	Tool / Version
Programming language	Python 3.12
Deep learning	PyTorch
Computer vision	OpenCV, scikit-image
Scientific computing	SciPy (elastic deformation, KD-tree)
Segmentation backbone	torchvision (DeepLabV3+)
Annotation / masking	ImageJ (Fiji distribution)
Hardware	NVIDIA T4 GPU, 16 GB VRAM

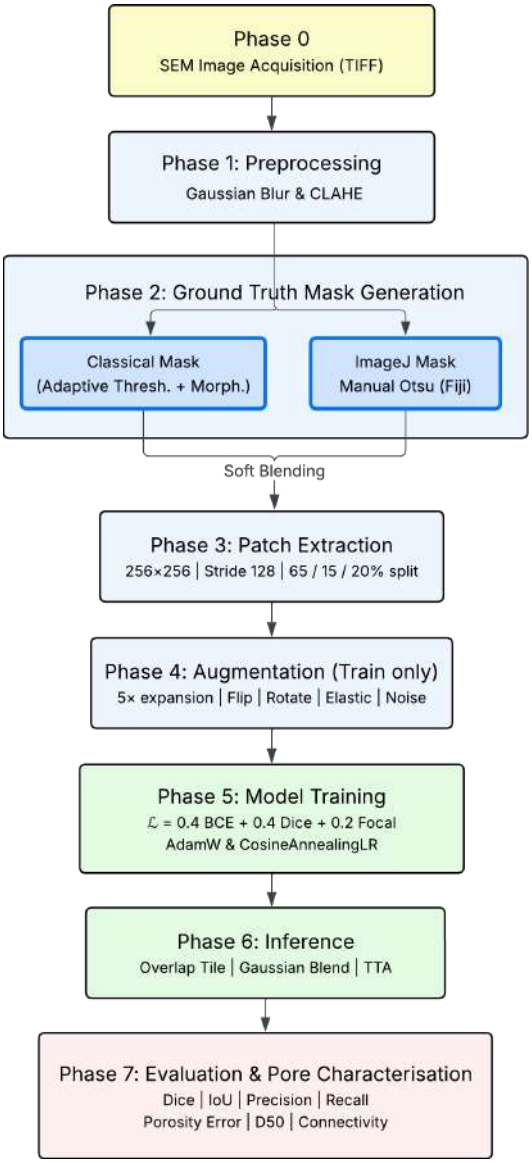


Figure 3.1: Seven-phase experimental pipeline shared across all five architectures.

3.2 SEM Dataset and Ground Truth Preparation

3.2.1 Image Acquisition and Dataset Composition

SEM images were acquired from seven porous ceramic specimens prepared at the Lab for Advanced Material Research (LAMR) by Md. Arif at Indian Institute of Technology (BHU) Varanasi. Each specimen was captured as a high-resolution 8-bit grayscale TIFF image. The dataset covers a broad porosity range (31%–48%) and exhibits diverse pore morphologies, ranging from isolated, nearly spherical pores to dense and irregularly interconnected pore networks. This variation ensures that the developed models learn to generalise across different microstructural features rather than simply memorising a single

pore geometry. The key properties of the dataset are summarised in *Table 3.2*.

Table 3.2: Dataset composition and patch statistics.

Property	Value
Image format	8-bit grayscale TIFF
Number of specimens	7
Porosity range	31%–48%
Classes	Pore / Matrix (binary)
Patch size	256×256 pixels
Total patches after filtering	≈ 378
Training patches after augmentation	1,225
Train / Validation / Test split	65% / 15% / 20%

The class imbalance between pore pixels (minority) and matrix pixels (majority) varies across specimens and is addressed at the loss-function level rather than by resampling, as described in Section 3.3.3.

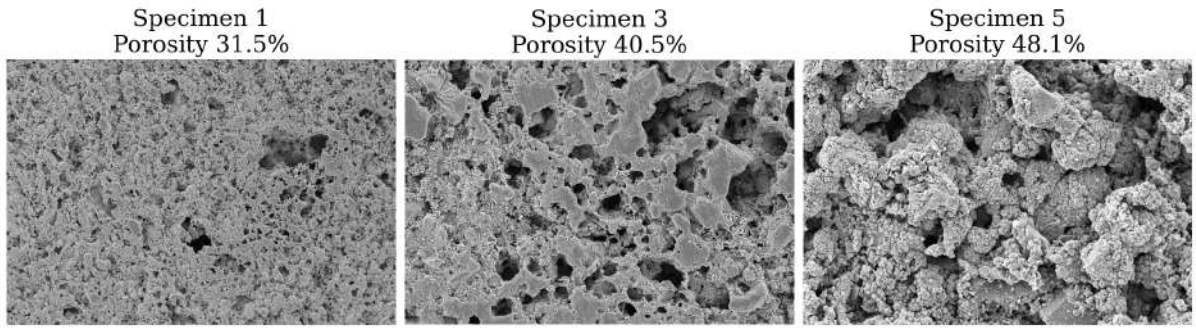


Figure 3.2: Representative SEM images across the porosity range of the seven specimens. Porosity values shown are ground-truth porosities derived from binarising the soft-blend mask at threshold 0.5, which is the pixel-wise union of the classical and ImageJ masks.

3.2.2 Image Preprocessing

Before mask generation and model training, all raw TIFF images undergo a two-step preprocessing pipeline. First, a **Gaussian blur** ($\sigma = 1$) is applied to reduce high-frequency sensor noise and charge-related pixel artefacts while preserving the pore boundaries that are critical for accurate segmentation.

Next, **CLAHE** (Contrast Limited Adaptive Histogram Equalisation; clip limit = 2.0, tile grid size = 8×8 px) is applied to improve local image contrast. In this process, the

image is divided into small tiles, each tile is equalised independently, and smooth bilinear interpolation is performed at the tile boundaries to avoid visible transitions. This step enhances pore-edge visibility in regions where the ceramic matrix compresses the overall grayscale distribution, while the clip limit helps prevent excessive amplification of noise in relatively uniform regions.

Both preprocessing operations were implemented using OpenCV (`cv2.createCLAHE`). The CLAHE-enhanced images, rather than the original raw TIFFs, were used for all subsequent stages, including mask generation and model training.

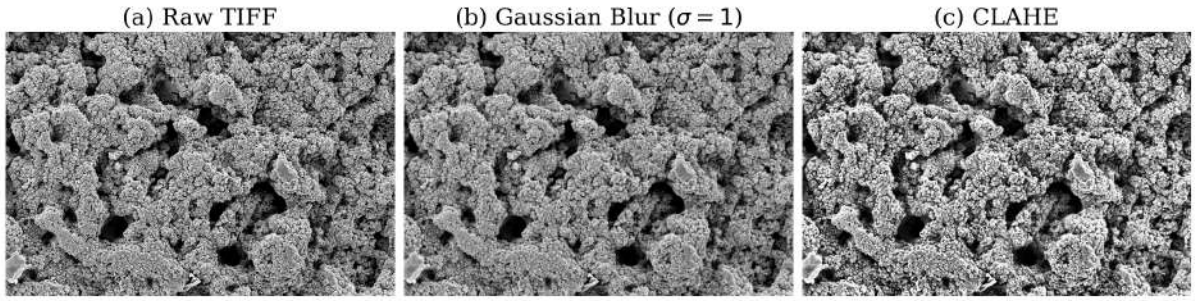


Figure 3.3: Preprocessing pipeline: raw TIFF → Gaussian blur → CLAHE.

3.2.3 Ground Truth Mask Generation

Because no single automated labelling method is consistently reliable on SEM pore images (limitations catalogued in *Table 1.1* of Chapter 1), ground-truth masks are produced by fusing two independently generated binary masks and computing formal inter-method agreement statistics.

Classical mask (automated via OpenCV + scikit-image). Applied to the CLAHE-enhanced image in five steps: (i) adaptive Gaussian thresholding (block = 31×31, $C = 5$, inverted so pore = white), the adaptive block accommodates SEM illumination gradients that render global Otsu thresholding unreliable; (ii) morphological opening (elliptical 3×3 kernel, 2 iterations) to remove thin noise bridges; (iii) morphological closing (same kernel, 2 iterations) to fill intra-pore gaps; (iv) removal of connected components < 100 px as sensor artefacts; (v) hole-filling for regions < 500 px to prevent pore fragmentation.

ImageJ mask (semi-manual via Fiji). For each specimen, a human analyst applied global Otsu thresholding in ImageJ (Fiji distribution) and saved the binary result as PNG. On import, the mask is inverted so that pore = white, consistent with the classical mask

convention. Crucially, the ImageJ and classical masks are produced entirely independently; no shared parameters, no analyst awareness of the automated result, ensuring genuine independence for the agreement check.

Soft-blend fusion. The two masks are fused by pixel-wise averaging:

$$M_{\text{final}}(x, y) = \frac{M_c(x, y) + M_j(x, y)}{2} \tag{3.1}$$

producing three confidence levels: 0 (neither method detects pore), 0.5 (one method agrees), 1 (both agree). This three-level encoding preserves inter-method uncertainty in the supervision signal rather than collapsing ambiguous boundary pixels to a hard label.

Inter-method agreement is quantified per specimen before any mask is accepted:

$$\text{IoU} = \frac{|M_c \cap M_j|}{|M_c \cup M_j|}, \qquad \text{Dice} = \frac{2|M_c \cap M_j|}{|M_c| + |M_j|}, \qquad \kappa = \frac{p_o - p_e}{1 - p_e} \tag{3.2}$$

where p_o is the observed pixel-level agreement fraction and p_e is the agreement expected by chance. Agreement results across all seven specimens are reported in Chapter 4.

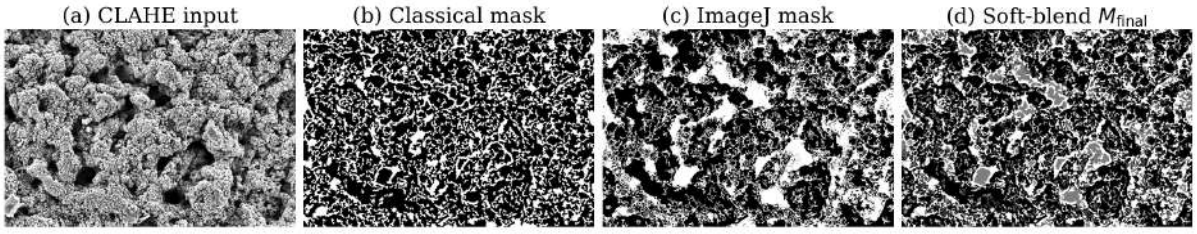


Figure 3.4: Ground-truth mask pipeline: CLAHE input, classical, ImageJ, and soft-blend fusion.

3.2.4 Patch Extraction, Splitting, and Augmentation

Patch extraction. Full-resolution CLAHE images are decomposed into 256×256 patches via a sliding window with stride 128 (50 % overlap). Patches where pore pixels account for fewer than 2 % of the area are discarded as uninformative background. The ≈378 retained patches are split at the patch level (not the specimen level) into training (65 %), validation (15 %), and test (20 %) subsets using a fixed random seed (42) for reproducibility. Patch-level splitting is used rather than specimen-level splitting because with only seven specimens a held-out specimen would represent approximately 14% of the data, making the split excessively coarse and leaving too few training samples for

reliable convergence. This choice introduces a data leakage caveat: overlapping patches from the same specimen can appear in both train and test subsets, which likely inflates the reported Dice and IoU figures relative to what would be observed on a fully unseen specimen. The reported metrics should therefore be treated as an upper bound on true generalisation performance; specimen-level evaluation is identified as the primary avenue for future work.

Augmentation. Training patches are expanded five-fold through the sequence below, applied identically to both the image patch and its mask. Augmentation is applied exclusively to the training set; validation and test sets are left unmodified.

1. Original
2. Horizontal flip
3. 90° rotation
4. Elastic deformation ($\alpha = 34$, $\sigma = 4$) via `scipy.ndimage.map_coordinates`, simulates the irregular boundary distortions of real pore networks.
5. Random brightness perturbation + additive Gaussian noise, simulating beam-current variation across SEM imaging sessions.

The 5× expansion brings the training set from ≈245 patches to 1,225, which is critical for TransUNet and DeepLabV3+, both of which carry more parameters than the lighter architectures and are more prone to overfitting on small datasets. *Figure 3.5* illustrates the five variants for one representative patch.

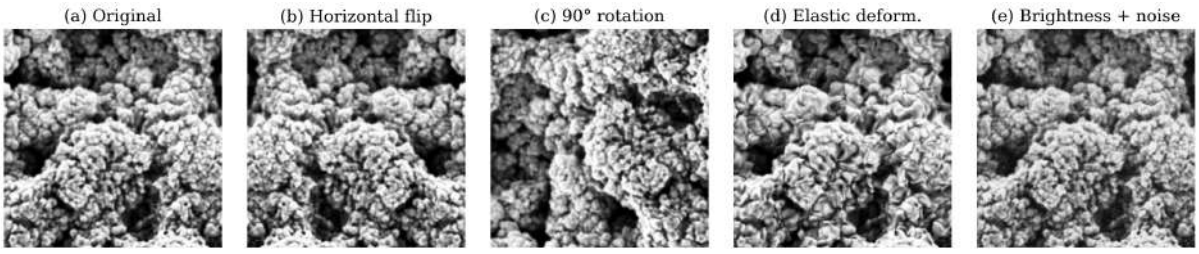


Figure 3.5: Five augmentation variants applied to one training patch.

3.3 Model Implementation and Training

3.3.1 Common Architecture Conventions

All five models were implemented in PyTorch and designed to process single-channel 256×256 grayscale input images. Each model outputs a single-channel logit map with the same spatial dimensions as the input, and a sigmoid activation function is applied to convert these logits into per-pixel pore probabilities.

To maintain architectural consistency and stable gradient propagation, SegNet, U-Net, Attention U-Net, and TransUNet all employ a shared residual double-convolution block consisting of two Conv–BN–ReLU layers combined with a 1×1 projection shortcut. The base number of feature channels was fixed at 32 for all custom models and doubled progressively at each encoder stage ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$). *Table 3.3* summarises the implementation parameters.

Table 3.3: Architecture hyperparameters for all five models (256×256×1 input, 32 base channels).

Model	Upsampling	Skip	Attn	ASPP	Bottleneck / Notes
SegNet	Pool-index unpool	No	No	No	No dropout
U-Net	Bilinear + conv	Full	No	No	Drop 0.4 / 0.15
DeepLabV3+	Bilinear (ASPP)	Low-lvl	No	Yes	ResNet-50; grayscale adapted; no pretrain
Attn U-Net	Bilinear + conv	Gated	Yes	No	Drop 0.4 / 0.15
TransUNet	Bilinear + conv	CNN	Yes	No	4 ViT layers; 4 heads; 256 tokens

3.3.2 Model-Specific Implementation Notes

SegNet

Five VGG-style encoder stages (2–3 Conv–BN–ReLU per stage) feed five symmetric decoder stages. The decoder unpools using the max-pooling indices recorded during the forward encoder pass (`nn.MaxUnpool2d`), placing activations back at their original spatial locations. No feature maps are forwarded across the encoder–decoder boundary beyond these indices. The model carries no dropout and is the lightest in the comparison.

U-Net

Four encoder stages each use the shared residual double-conv block, separated by max-pooling. The decoder bilinearly upsamples and concatenates the corresponding full encoder feature map before another residual conv. Dropout of 0.4 is applied at the bottleneck and 0.15 at the two upper decoder levels to reduce overfitting without over-regularising the boundary-sensitive upper decoder.

DeepLabV3+

DeepLabV3+ was implemented using the `torchvision` framework with a ResNet-50 backbone. Since the SEM images are single-channel grayscale inputs, the first convolutional layer of the backbone was modified to accept a single input channel instead of the standard three-channel RGB format. All network weights were trained from scratch without using ImageNet pretraining.

The model employs Atrous Spatial Pyramid Pooling (ASPP), which combines parallel dilated convolutions with dilation rates of $\{6, 12, 18\}$ together with a global average pooling branch to capture multi-scale contextual information.

To improve training stability, a custom `SafeBatchNorm` wrapper was introduced. This modification prevents failures in the global pooling branch when the spatial resolution collapses to 1×1 during the forward pass by temporarily switching Batch Normalisation to evaluation-mode statistics under those conditions.

Attention U-Net

Architecturally identical to U-Net, with one addition: each skip connection passes through an Attention Gate (AG) before concatenation. The gate computes a spatial attention map $\alpha \in (0,1)^{H \times W}$ from the encoder feature $\mathbf{x}_l$ and the upsampled decoder signal $\mathbf{g}$:

$$\alpha = \sigma\big(\mathbf{W}_\psi \,\mathrm{ReLU}(\mathbf{W}_g\,\mathbf{g} + \mathbf{W}_x\,\mathbf{x}_l)\big) \tag{3.3}$$

The gated feature $\alpha \odot \mathbf{x}_l$ suppresses matrix activations and amplifies pore-boundary signals prior to concatenation. Implemented as three 1×1 convolutions with negligible parameter overhead.

TransUNet

After four convolutional downsampling stages, the resulting 16×16 feature map is projected into 512 channels and reshaped into a sequence of 256 tokens. This token sequence is then processed by four pre-normalised Transformer encoder layers, each containing 4 multi-head

self-attention heads, an MLP expansion ratio of 4, and a dropout rate of 0.1.

Through the self-attention mechanism, every spatial location in the feature map can interact with every other location in the image. This enables the model to capture long-range pore interconnectivity and global structural relationships that are difficult for conventional CNN-based architectures to model effectively.

After Transformer processing, the tokens are reshaped back into a 16×16 spatial feature grid and progressively decoded using bilinear upsampling along with CNN skip connections from the four encoder stages.

3.3.3 Training Strategy and Hyperparameter Configuration

All five models are trained under the identical configuration in *Table 3.4*; no model-specific tuning is applied.

Table 3.4: Training configuration (identical for all five models).

Parameter	Value
Loss function	$\mathcal{L} = 0.4\,\mathcal{L}_{\text{BCE}} + 0.4\,\mathcal{L}_{\text{Dice}} + 0.2\,\mathcal{L}_{\text{Focal}}(\gamma{=}2)$
Class balancing	$w_+ = N_{\text{bg}}/N_{\text{pore}}$ applied to BCE
Optimiser	AdamW
Learning rate	1×10^{-3}
LR schedule	CosineAnnealingLR ($T_{\max} = 150$)
Batch size	8
Max epochs	150
Early stopping	Patience = 25 on validation Dice
Checkpointing	Save best val. Dice; auto-resume every 10 epochs

Loss function rationale. The three loss terms address complementary failure modes. BCE ($\mathcal{L}_{\text{BCE}}$) provides per-pixel probability calibration and a stable gradient throughout training. Dice loss ($\mathcal{L}_{\text{Dice}}$) directly optimises the overlap metric reported in Chapter 4 and handles class imbalance intrinsically:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon} \tag{3.4}$$

where p_i and g_i are the predicted probability and ground-truth label at pixel i, and $\varepsilon = 1$ prevents division by zero. Focal loss ($\mathcal{L}_{\text{Focal}}$, $\gamma = 2$) down-weights confidently correct

background predictions and concentrates the gradient on hard cases like pore boundaries, small isolated pores, and low-contrast regions. AdamW is preferred over standard Adam because its decoupled weight decay is more consistent across the heterogeneous layer types (convolutional, batch-norm, self-attention) present in the model set. CosineAnnealingLR smoothly decays the learning rate to near-zero, avoiding the gradient instability that step-decay schedules can introduce in Transformer layers.

3.4 Inference, Evaluation, and Pore Characterisation

3.4.1 Overlap-Tile Inference

Full SEM images exceed the 256×256 training patch size, so inference uses an overlap-tile strategy that mirrors patch extraction exactly. The image is scanned with a 256×256 window at stride 128; each tile is independently forwarded to yield a probability map. A 2-D Gaussian kernel (σ = patch size/4) down-weights each tile's edge predictions, where limited context degrades accuracy. Weighted tile outputs are accumulated on a full-image canvas and normalised by the summed Gaussian weight at each pixel, eliminating stitching artefacts. Test-Time Augmentation (TTA) additionally predicts each tile horizontally flipped, un-flips the output, and averages it with the original before Gaussian weighting. The final probability map is thresholded at 0.5 to yield the binary pore mask.

3.4.2 Quantitative Evaluation

Per-specimen segmentation performance is measured against the binarised soft-blend ground truth using Dice, IoU, Precision, Recall, and Pixel Accuracy (defined in Chapter 2, Section 2.5). One additional metric is reported here: **Porosity Error** is the absolute difference $|\hat{\phi} - \phi_{\mathrm{GT}}|$ between predicted and ground-truth porosity fractions, which directly quantifies the functional consequence of segmentation bias for the Gibson–Ashby structure–property relationship. All five models are evaluated on the same held-out test set and inference protocol.

3.4.3 Qualitative Evaluation

Predicted masks are overlaid on the original SEM images and inspected for: boundary smoothness and continuity along pore edges; false positives (predicted pore regions in

matrix areas); false negatives (missed pores, particularly small or low-contrast ones); and topological consistency of connected pore networks. Qualitative assessment complements the quantitative metrics because two models with similar Dice scores can produce visually distinct segmentation maps. Per-specimen overlay figures are included in Chapter 4.

3.4.4 Pore Characterisation

Following binary segmentation by the best-performing model, connected-component labelling (`skimage.measure.label`) identifies individual pore objects. scikit-image `regionprops` extracts per-pore shape descriptors and SciPy's `KDTree` computes spatial proximity statistics. *Table 3.5* lists the precise computational definitions used, extending the morphological descriptors introduced in *Table 2.1*.

Table 3.5: Pore characterisation descriptors from connected-component analysis.

Descriptor	Definition
Equivalent diameter	$d = \sqrt{4A/\pi}$; circle diameter with equal area A
D10 / D50 / D90	10th, 50th, 90th percentile of equivalent diameter distribution
Circularity	$4\pi A/P^2$ ($= 1$ for perfect circle)
Aspect ratio	Major axis / minor axis of fitted ellipse
Solidity	A/A_{convex}; low values indicate concave pores
Eccentricity	Eccentricity of fitted ellipse (0 = circle, 1 = line)
Nearest-neighbour dist.	Mean Euclidean distance to closest pore centroid (KD-tree)
Euler number	#components − #holes; more negative ⇒ more connected
Connectivity ratio	|Euler no.| / N_{pore}; normalised connectivity

Chapter 4

Results and Discussion

4.1 Ground-Truth Agreement

Before evaluating any model, the quality of the soft-blend ground truth must be established. *Table 4.1* reports the inter-method agreement between the classical OpenCV mask and the ImageJ Fiji mask for each specimen, using the IoU, Dice, and Cohen's κ defined in Chapter 3.

Table 4.1: Inter-method agreement between classical and ImageJ masks per specimen. †GT ϕ is the porosity of the binarised soft-blend mask (threshold ≥ 0.5, equivalent to the pixel-wise union of both masks); this is the ground truth used for model training and evaluation.

Specimen	IoU	Dice	κ	Classical ϕ (%)	ImageJ ϕ (%)	GT ϕ (%)†
1	0.509	0.674	0.575	27.2	20.3	31.5
2	0.453	0.624	0.482	30.8	24.7	38.2
3	0.457	0.627	0.477	34.0	24.9	40.5
4	0.571	0.727	0.574	33.5	38.8	46.0
5	0.524	0.687	0.507	35.3	38.0	48.1
14	0.559	0.717	0.598	32.1	27.4	38.1
15	0.584	0.737	0.647	28.1	23.6	32.6
Mean	0.522	0.685	0.551	31.6	28.2	39.3

While referring to *Figure 4.1*, agreement ranges from moderate to substantial (κ 0.48–0.65), which is expected given that the two methods apply fundamentally different thresholding assumptions to images with gradient illumination. Importantly, the methods *disagree systematically* rather than randomly: classical thresholding consistently over-segments low-porosity specimens whereas ImageJ under-segments them, and the relationship reverses at higher porosity. This systematic divergence motivates the soft-blend fusion strategy. The soft-blend mask is formed by pixel-wise averaging the two binary masks, yielding values of 0 (neither detects pore), 0.5 (one method detects pore),

or 1.0 (both agree). Binarising at a threshold of 0.5 is equivalent to taking the pixel-wise union of both masks, and it is this binarised union that defines the ground-truth porosity (GT ϕ, last column of *Table 4.1*) used for all model training and evaluation. The union convention means GT ϕ values (31% to 48%) are higher than either method individually. The float-valued soft mask is retained during training to convey boundary uncertainty at pixels where only one method detected a pore.

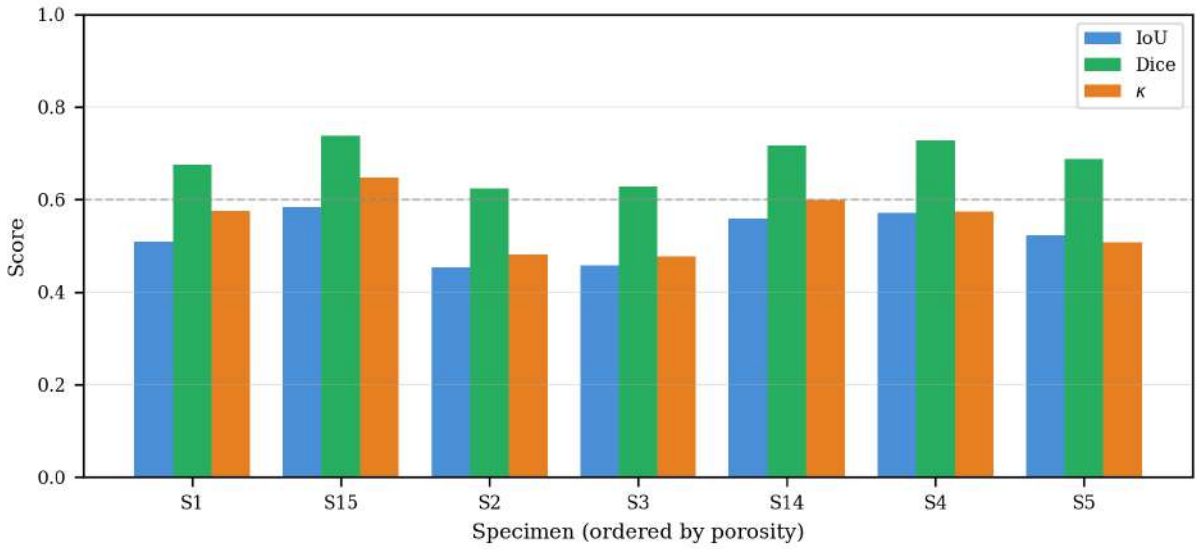


Figure 4.1: Per-specimen inter-method agreement (specimens ordered by final porosity).

4.2 Segmentation Benchmark

4.2.1 Overall Metrics

Table 4.2 summarises the five-model benchmark on the held-out test set. All models are evaluated under the identical overlap-tile inference protocol described in Chapter 3.

Table 4.2: Overall test-set metrics for the five architectures (pp = percentage points).

Model	Dice	IoU	Prec.	Rec.	Porosity Err.
Attention U-Net	**0.983**	**0.967**	0.981	**0.985**	0.17 pp
U-Net	0.982	0.965	**0.982**	0.982	**0.05 pp**
TransUNet	0.970	0.941	0.962	0.977	0.61 pp
SegNet	0.950	0.905	0.942	0.959	0.73 pp
DeepLabV3+	0.869	0.769	0.844	0.895	2.39 pp

Figure 4.2 shows three performance tiers clearly.

Tier 1 (Attention U-Net, U-Net): Dice ≥ 0.982, near-identical boundary precision, negligible porosity error. The attention gates in Attention U-Net suppress false positives in matrix-rich regions, yielding marginally higher recall without sacrificing precision. U-Net's simpler full-feature skip connections achieve comparable overlap but introduce a slight recall–precision asymmetry.

Tier 2 (TransUNet, SegNet): Dice 0.950–0.970. TransUNet's global self-attention is architecturally powerful but underutilised on a 7-specimen dataset where long-range pore inter-connectivity is not the binding constraint; its heavier parameter count also increases its regularisation burden. SegNet's index-only skip connections discard fine feature maps, limiting boundary sharpness at finer pore edges.

Tier 3 (DeepLabV3+): Dice 0.869, porosity error 2.39 pp. Training the ResNet-50 backbone from scratch on a single-channel grayscale input, without ImageNet pretraining, limits what the model can learn from 1,225 patches; the ASPP multi-scale pooling adds representational capacity that this dataset size cannot fully saturate. Although DeepLabV3+ has previously outperformed U-Net on larger multi-scale pore datasets reported in the literature [15], its weaker performance here is attributable to the limited dataset size, the grayscale single-channel adaptation, and the absence of pretrained weights. Architectures with deep residual backbones and ASPP generally require substantially larger and more diverse training sets for stable optimisation.

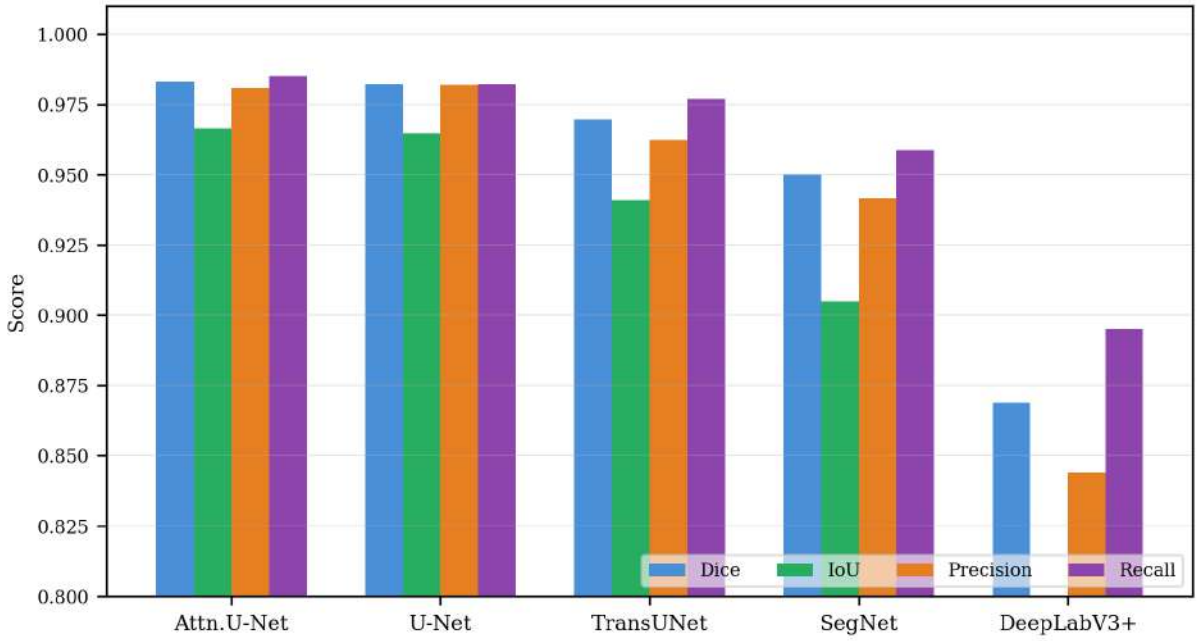


Figure 4.2: Model comparison across Dice, IoU, Precision, and Recall on the held-out test set.

4.2.2 Per-Specimen Breakdown

The heatmap represented in *Figure 4.3* reveals two consistent patterns. First, all models score higher on denser, mid-to-high porosity specimens (3, 5), where pores form clear, contiguous regions and the contrast between pore and matrix is strongest. Second, DeepLabV3+ is the only model to exhibit specimen-level Dice below 0.85, confirming that its shortfall is systematic rather than specific to any morphology type.

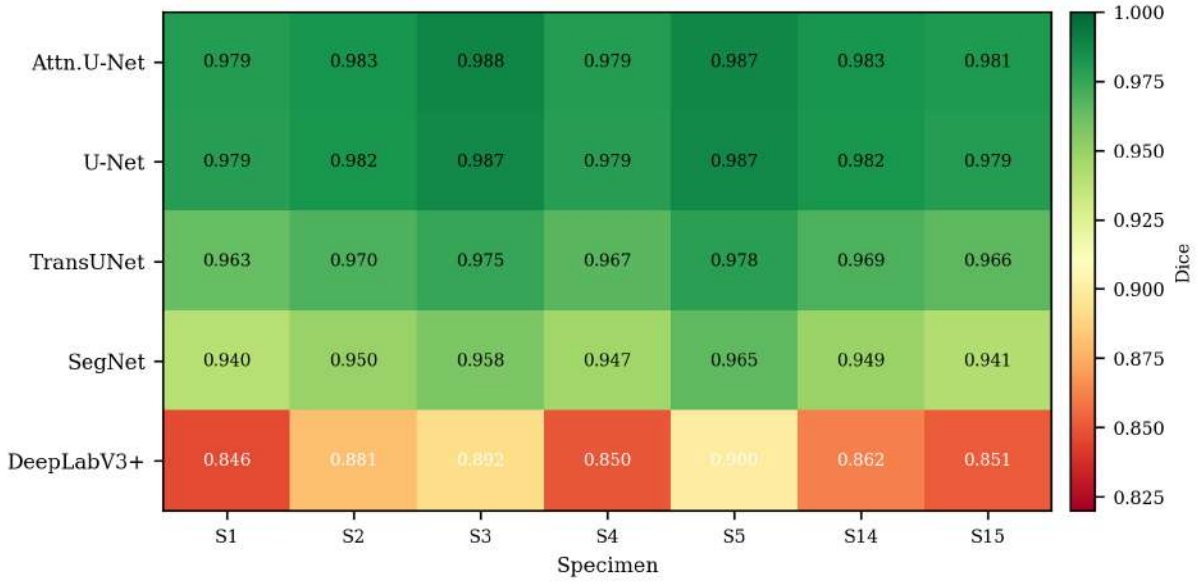


Figure 4.3: Per-specimen Dice for all five models (specimens ordered by ID).

4.2.3 Porosity Estimation Accuracy

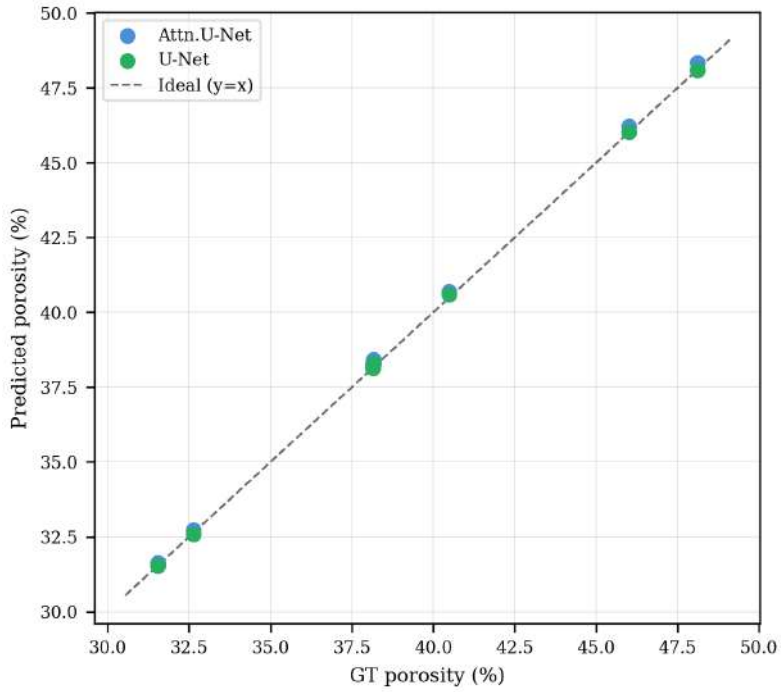


Figure 4.4: Predicted vs. ground-truth porosity for Attention U-Net and U-Net; dashed line is ideal ($y = x$).

Porosity is derived directly from the predicted binary mask rather than from any regression model. Attention U-Net achieves a mean absolute porosity error of 0.17 percentage points across the seven specimens; U-Net achieves 0.04 pp, a near-zero error that confirms the segmentation mask is a faithful quantitative proxy for specimen porosity. Both models lie visually on the $y = x$ line, indicating that neither method introduces a systematic bias toward over- or under-segmentation across the 31%–48% porosity range.

4.3 Pore Characterisation

4.3.1 Specimen Gallery and Predicted Properties

Figure 4.5 shows the CLAHE image alongside the predicted connected-component map for five specimens, with key morphological descriptors computed from the Attention U-Net segmentation (best model).

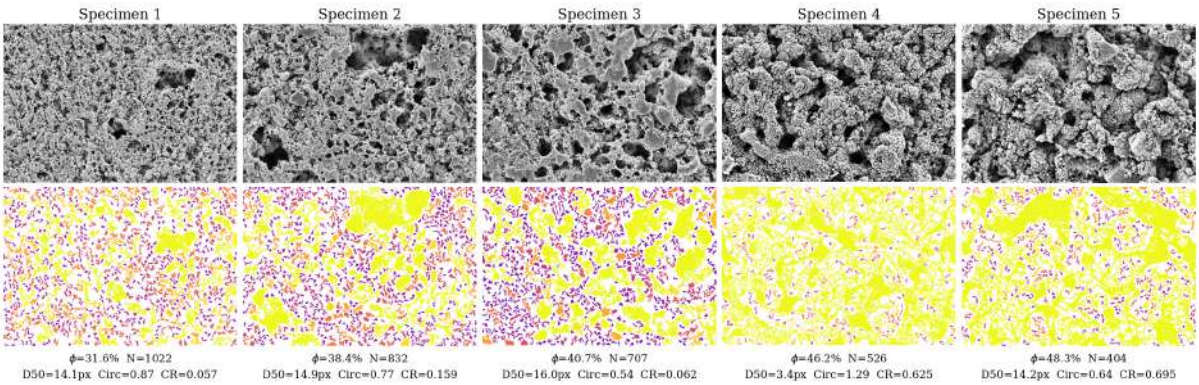


Figure 4.5: Per-specimen gallery: CLAHE input (top) and predicted component map coloured by pore diameter (bottom). Reported descriptors are derived from the Attention U-Net predicted mask: predicted porosity ϕ, pore count N, median diameter D50, solidity, aspect ratio AR, and connectivity ratio CR.

The colour-coded component maps reveal the structural diversity of the dataset at a glance: low-porosity specimens (1, 2) display sparse, well-separated pores rendered in cooler tones that reflect smaller equivalent diameters, whereas high-porosity specimens (4, 5) show densely packed, larger regions with warmer hues and visibly higher connectivity. These qualitative contrasts are captured numerically by the full descriptor set, which *Table 4.3* reports across all seven specimens.

Table 4.3: Pore descriptors per specimen derived from Attention U-Net predictions: predicted porosity ϕ (GT values in Table 4.1), pore count N, median diameter D50, solidity, aspect ratio AR, and connectivity ratio CR.

Spec.	ϕ (%)	N	D50 (px)	Solid.	AR	CR
1	31.6	1022	14.1	0.749	1.93	0.057
2	38.4	832	14.9	0.747	2.00	0.159
3	40.7	707	16.0	0.728	2.04	0.062
4	46.2	526	3.4	0.796	1.82	0.625
5	48.3	404	14.2	0.720	2.05	0.695
14	38.3	661	13.6	0.726	1.94	0.066
15	32.7	845	13.5	0.742	1.90	0.027

4.3.2 Morphological Trends

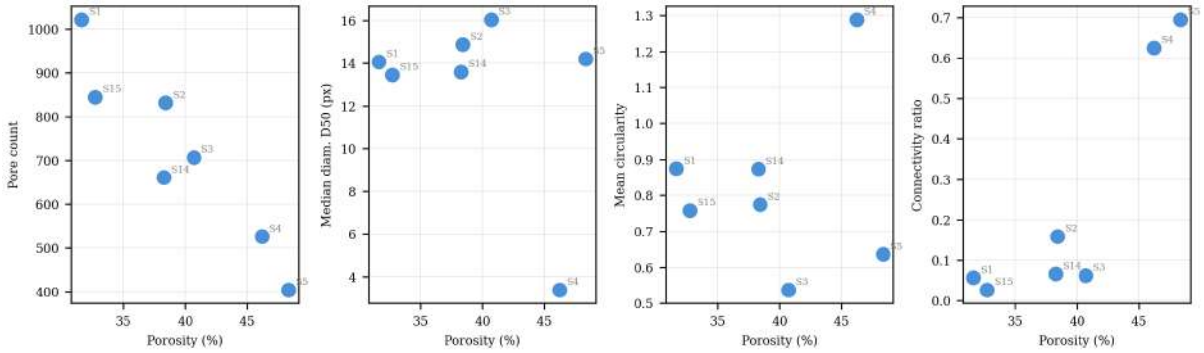


Figure 4.6: Pore-network descriptors vs. porosity across all seven specimens.

Four trends emerge from *Figure 4.6* and *Table 4.3*:

1. **Pore count decreases with porosity.** Specimens at 31–33% porosity contain 845–1022 discrete pores; at 46–48% porosity, only 404–526 remain identifiable as separate objects. Higher sintering temperatures promote pore coalescence, reducing object count while inflating individual pore volume.
2. **Connectivity ratio increases sharply between 41% and 46% porosity.** Specimens at or below 40.7% porosity all have CR ≤ 0.16; specimens 4 and 5 (46% and 48%) reach CR $= 0.625$ and 0.695, respectively. The data bracket the transition between specimen 3 (40.7%, CR $= 0.062$) and specimen 4 (46.2%, CR $= 0.625$), consistent with the onset of a percolating pore network expected near the classical percolation threshold for sintered porous ceramic microstructures.
3. **Specimen 4 exhibits a distinct microstructural character.** Despite having the second-highest porosity (46.2%), its median pore diameter D50 is only 3.4 px, markedly smaller than the 13–16 px range seen across the other specimens. One possible explanation is that the fine pore structure results from processing conditions that suppressed coalescence: lower sintering temperature or shorter dwell time limits the grain-boundary migration needed to merge adjacent voids, leaving a large number of small, unmerged pores [12]. Starting powder particle size and pore-former geometry are alternative variables that influence pore size independently, and without processing records for these specimens the underlying mechanism cannot be confirmed. Given the sub-5 px median pore size, other morphological descriptors for this specimen should be treated with caution relative to the remaining six.

4. **Aspect ratio is largely invariant below 40% porosity**, with AR $\approx$ 1.90–2.00. Above 40%, AR rises slightly, consistent with elongated, channel-like pore morphology at higher void fractions.

4.3.3 Qualitative Assessment

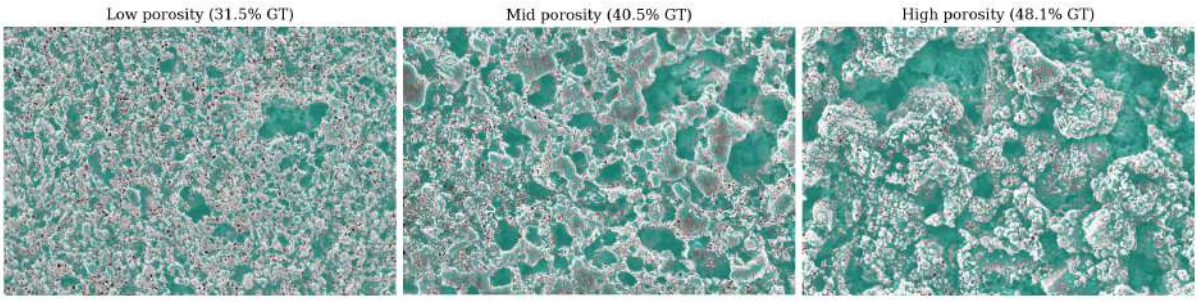


Figure 4.7: Predicted pore regions (teal overlay) on CLAHE-enhanced SEM images at low, mid, and high porosity.

The overlay confirms spatially coherent segmentations that respect pore boundaries across the full porosity range, with no systematic over- or under-segmentation at any pore scale.

The descriptors together characterise each specimen as a compact vector comparable across sintering conditions or regressable against macroscopic properties such as strength and permeability: evidence that the gap between a raw SEM image and a quantitative microstructural description can be closed reliably with a well-trained attention-based segmentation model.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This thesis presented a comparative benchmark for automated pore segmentation in SEM images of porous ceramics, focusing on segmentation accuracy, ground-truth generation, and quantitative pore-network analysis. The key findings are:

1. **Attention U-Net achieved the best overall performance** (Dice 0.983, IoU 0.967), closely followed by U-Net, demonstrating that attention-enhanced encoder–decoder architectures are highly effective for ceramic pore segmentation.
2. **Soft-blend ground-truth masks improved supervision quality.** Moderate disagreement between classical and ImageJ masks justified averaging both annotations to better preserve pore-boundary uncertainty.
3. **Segmentation enabled accurate porosity estimation.** U-Net and Attention U-Net predicted porosity with very low mean absolute errors, validating the framework as an automated alternative to manual ImageJ analysis.
4. **Transformer and ASPP-based models underperformed** due to the limited dataset size, with DeepLabV3+ showing the weakest convergence and highest porosity error.
5. **A pore-network transition was observed between 41% and 46% porosity,** where isolated pores evolved into interconnected channel-like structures, with the connectivity ratio rising sharply between specimen 3 (40.7%, CR = 0.062) and specimen 4 (46.2%, CR = 0.625).

Compact encoder–decoder networks with attention proved sufficient for accurate, fully automated pore segmentation on this dataset, and the extracted descriptors capture real microstructural variation across sintering conditions. Given that the train/test split was performed at the patch level rather than the specimen level, the reported Dice and IoU figures should be interpreted as an optimistic upper bound on generalisation performance;

a specimen-level held-out evaluation would provide a more conservative and reliable estimate. The comparative performance trends reported in this study are inherently dependent on the size, diversity, and characteristics of the available SEM dataset. Since the models were trained and evaluated on a limited dataset comprising seven porous ceramic specimens, the reported accuracy metrics should not be interpreted as universally fixed performance indicators for these architectures. Variations in dataset scale, pore morphology distribution, imaging conditions, annotation quality, or material system may alter the relative ranking and segmentation accuracy of the evaluated models. In particular, architectures such as DeepLabV3+ and TransUNet generally benefit from substantially larger and more diverse training datasets, and their performance may improve under broader data availability.

5.2 Limitations

The main limitation of this work is the small dataset size, which restricts the generalisation capability of larger architectures. In addition, the 2-D SEM analysis cannot fully capture fine-scale or true 3-D pore-network topology.

5.3 Future Work

Future work may focus on:

1. Expanding the dataset across more specimens and material conditions to improve generalisation.
2. Linking pore descriptors with mechanical and thermal properties through regression models.
3. Extending the framework to 3-D pore reconstruction using FIB-SEM data.
4. Exploring semi-supervised and few-shot learning to reduce annotation dependence.
5. Developing lightweight real-time segmentation for direct SEM workstation deployment.
6. Investigating ensemble strategies (e.g. weighted averaging of Attention U-Net, U-Net, and TransUNet) to determine whether combined predictions can push accuracy beyond any single model on this dataset.

Additional Data and Figures

This section provides a full per-specimen breakdown of the segmentation results produced by the best-performing model (Attention U-Net). Each figure shows, left to right: the raw TIFF image, the CLAHE-enhanced input, the soft-blend ground-truth mask, and the predicted pore overlay. Specimen identifiers, ground-truth and predicted porosity, Dice score, and IoU are noted beneath each figure.

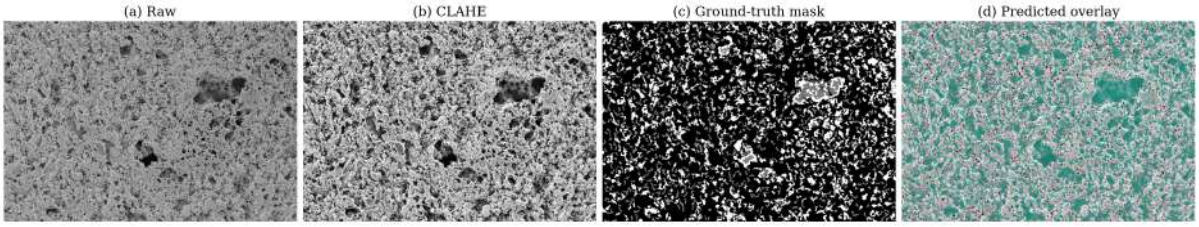


Figure A.1: Specimen 1 (GT porosity 31.5%, predicted 31.6%, Dice 0.979).

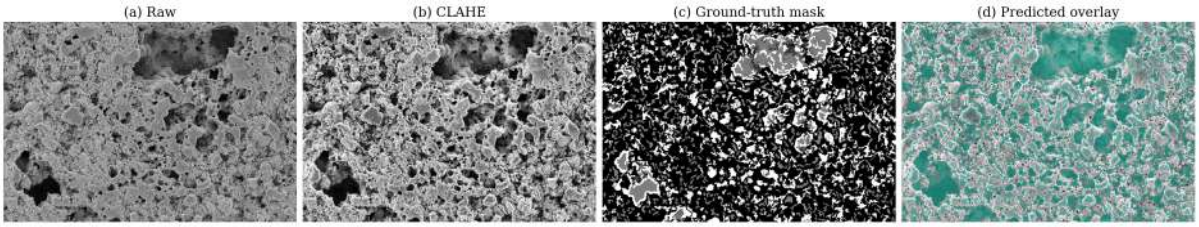


Figure A.2: Specimen 2 (GT porosity 38.2%, predicted 38.4%, Dice 0.983).

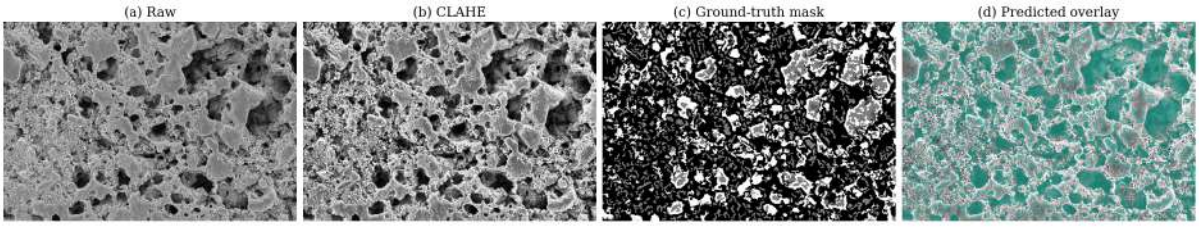


Figure A.3: Specimen 3 (GT porosity 40.5%, predicted 40.7%, Dice 0.988).

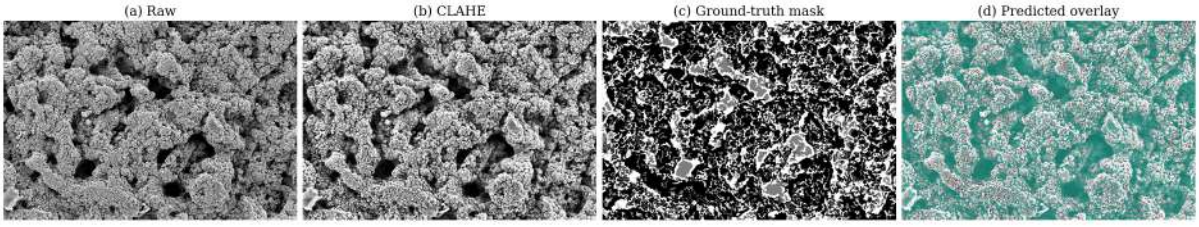


Figure A.4: Specimen 4 (GT porosity 46.0%, predicted 46.2%, Dice 0.979).

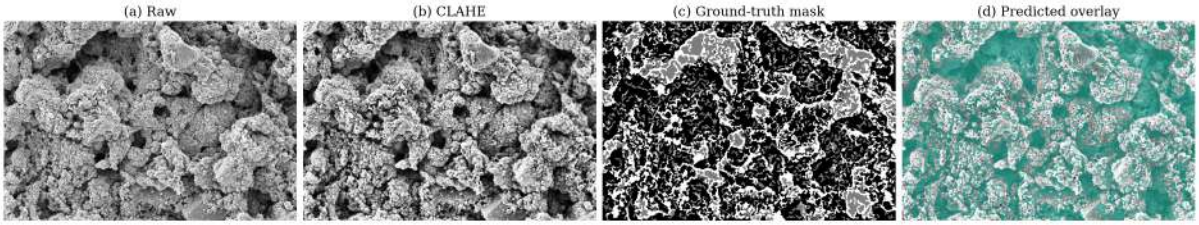


Figure A.5: Specimen 5 (GT porosity 48.1%, predicted 48.3%, Dice 0.987).

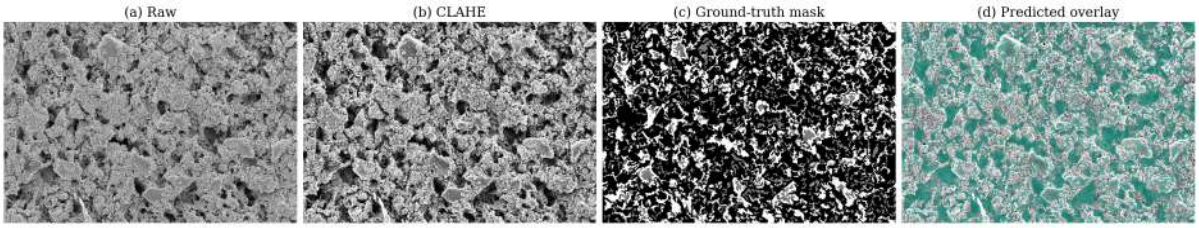


Figure A.6: Specimen 14 (GT porosity 38.1%, predicted 38.3%, Dice 0.983).

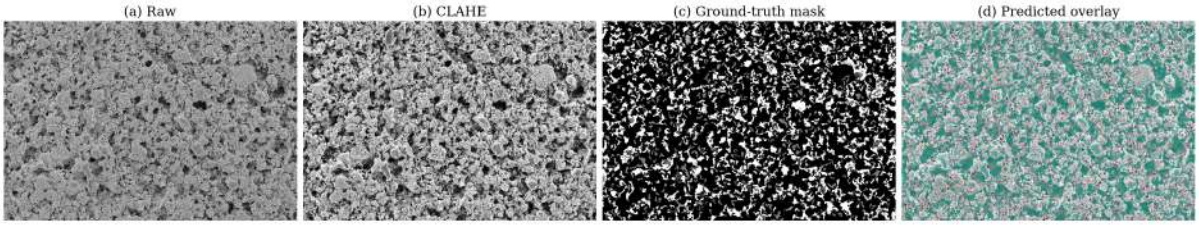


Figure A.7: Specimen 15 (GT porosity 32.6%, predicted 32.7%, Dice 0.981).

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